Observer Design for a Class of Nonlinear Port-Hamiltonian Systems

Master's Thesis Defense

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SCIENCES & TECHNOLOGIES

Outline

Introduction and Motivation

Mechanical and Electromechanical Models

Structure-Preserving Observer via Contraction

Literature result

Thesis Contributions

Application to HASEL Actuators

Introduction and Motivation

Context: Electromechanical Soft Actuators

Motivation:

- Emerging applications in **soft robotics** (HASEL, DEA) and medical devices
- Multi-physics couplings lead to **strongly nonlinear dynamics**
- Some internal states **not directly measurable** $\Rightarrow$ reconstruction essential for IDA-PBC

Problem Statement

How to reconstruct internal states while preserving the physical structure and passivity properties of the port-Hamiltonian model?

References: HASEL [1], [2] • DEA [3], [4]

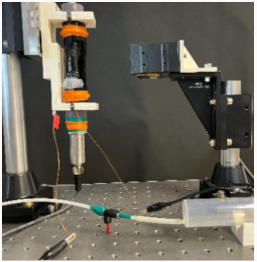


Figure 1: DEA actuator

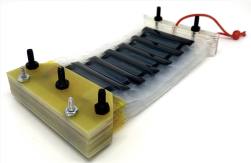


Figure 2: HASEL actuator

Port-Hamiltonian (PH) Framework

General form of a pH system [5]

$$\dot{x} = [J(x) - R(x)]\nabla H(x) + g(x)u, \quad y = g(x)^\top \nabla H(x)$$

- $x \in \mathbb{R}^n$: state vector (energy variables),
- $H(x)$: Hamiltonian function (total energy),
- $J(x) = -J(x)^\top$: skew-symmetric interconnection matrix,
- $R(x) = R(x)^\top \geq 0$: symmetric dissipation matrix,
- $g(x)$: input matrix,
- y: collocated output (power-conjugate).

Key properties:

- ✔ Explicit physical structure
- ✔ Passivity
- ✔ Unified framework for multi-physics systems

Mechanical and Electromechanical Models

Mechanical PH Model

State variables: $x = [q, p]^\top$ (displacement, momentum)

Hamiltonian ($H_m(x)$) and Hamiltonin gradient ($\nabla H_m(x)$)

$$H_m(x) = \frac{1}{2}Kq^2 + \frac{p^2}{2m} \implies \nabla H_m(x) = \begin{bmatrix} Kq \\ p/m \end{bmatrix} = Qx, \quad Q = \begin{bmatrix} K & 0 \\ 0 & 1/m \end{bmatrix}$$

Considering $u(t) = U(t)^2$, the dynamic can be written as follows

$$\begin{bmatrix} \dot{q} \\ \dot{p} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -1 & -\eta \end{bmatrix}}_{J-R} \begin{bmatrix} Kq \\ p/m \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 2\varepsilon(q+q_0)^3 \end{bmatrix}}_{g(x)} u(t)$$

Key feature

Nonlinearity only in $g(x)$, constant J, R, quadratic H.

Electromechanical PH Model

State variables: $x = [q, p, Q]^\top$ (displacement, momentum, charge)

Hamiltonian ($H_e(x)$) and Hamiltonin gradient ($\nabla H_e(x)$)

$$H_e(x) = \frac{1}{2}Kq^2 + \frac{p^2}{2m} + \frac{Q^2}{2C(q)} \implies \nabla H_e(x) = \begin{bmatrix} Kq - \frac{2Q^2}{\varepsilon(q+q_0)^5} \\ p/m \\ \frac{Q}{\varepsilon(q+q_0)^4} \end{bmatrix}$$

Considering $C(q) = \varepsilon(q + q_0)^4$, the dynamic can be written as follows

$$\begin{bmatrix} \dot{q} \\ \dot{p} \\ \dot{Q} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & -\eta & 0 \\ 0 & 0 & -\frac{1}{r_e} \end{bmatrix} \nabla H_e(x) + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{r_e} \end{bmatrix} u(t)$$

Key feature

Constant J, R, g, but nonlinear $H(x)$

Structure-Preserving Observer via Contraction

Observer Structure: general case

$$\begin{cases} \dot{x} = [J(x) - R(x)]\nabla H(x) + g(x)u, \quad y = g(x)^\top \nabla H(x), \\ \dot{\hat{x}} = [J(\hat{x}) - R(\hat{x})]\nabla H(\hat{x}) + g(\hat{x})u + L(\hat{x})(\hat{y} - y), \quad \hat{y} = g(\hat{x})^\top \nabla H(\hat{x}). \end{cases}$$

By substitution of the collocated outputs in the observer dynamic the following equivalent form is obtained

$$\dot{\hat{x}} = [J(\hat{x}) - R(\hat{x}) - L(\hat{x})g(\hat{x})]\nabla H(\hat{x}) + g(\hat{x})u + L(\hat{x})g(x)\top\nabla H(x).$$

Key Property

The observer retains the energy-based interpretation and the output injection acts through the same channel as the passive output.

Challenge

Prove the exponential convergence of the error dynamics, to do so is possible to exploit the contraction theory.

Contraction Theory [6], [7]

Core Philosophy

Analyze **distance between trajectories**, not just equilibrium stability

Nonlinear system $\dot{x} = f(x,t), \quad x \in \mathbb{R}^n, \quad f \in \mathcal{C}^1$,

Variational dynamics: $\delta\dot{x} = \underbrace{\frac{\partial f}{\partial x}(x,t)}_{A(x,t)} \delta x$,

Lyapunov function: $V(x, \delta x, t) = \delta x^\top M(x,t) \delta x$.

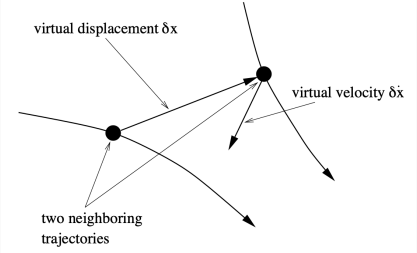


Figure 3: Virtual displacement δx between neighboring trajectories.

Contraction Condition

System is **contracting** with rate $\lambda > 0$ if $\exists$ metric $M(x,t) = M(x,t)^\top \succ 0$:

$$\dot{M} + A^\top M + MA \preceq -2\lambda M$$

Contraction Condition: Properties and Special Cases

Consequence: Incremental Exponential Stability

Distance between *any* pair of trajectories decreases exponentially:

$$\|\delta x(t)\|_M \leq e^{-\lambda t}\|\delta x(0)\|_M \quad \Rightarrow \quad \text{all trajectories converge to each other}$$

Special Case: Constant Metric

If $M(x,t) = P = P^\top \succ 0$ (constant), the contraction condition simplifies to:

$$A(x,t)^\top P + PA(x,t) \preceq -2\lambda P$$

This resembles the Lyapunov matrix inequality for linear systems

Key Distinction: Contraction analyzes trajectory convergence
Lyapunov stability analyzes convergence to equilibrium

Literature result

Existing Contraction-Based Approaches [8]

Literature results

Constant output injection gain L and input matrix $g(x) = g$.

Case A: Nonlinear interconnection/dissipation

- Quadratic Hamiltonian $\Rightarrow$ linear $\nabla H(x) = Qx$,
- Nonlinear $J(x)$ and/or $R(x)$.

Case B: Nonlinear Hamiltonian

- Nonquadratic Hamiltonian $\Rightarrow$ nonlinear $\nabla H(x)$, presence of coupling within the difference dynamics,
- Constant J and R.

Verification method: It is proven that exponential convergence is guaranteed by checking the passive inequality with the Hessian as contraction metric: $R(x) + \mathrm{sym}(gL^\top) \geq \beta I \succ 0$.

Limitations of Contraction-Based Design

⚠ Three Key Limitations

1. ⊘ **Restrictive structural assumptions:**
   - Consider only constant L (no gain scheduling),
   - Nonlinearities in the input matrix are not considered,
   - Even for Cases A & B systems, passive inequality may not be satisfied

2. ✖ **Verification difficulties:**
   - Finding suitable metric is non-trivial
   - For highly coupled systems, checking principal minors of contraction inequality is challenging

3. ❓ **No systematic synthesis:**
   - Manual trial-and-error for metric selection
   - No optimization framework for gain L

→ Need for a more general and systematic approach [9]

Thesis Contributions

LPV Formulation - Case 1: Nonlinear Input Matrix

System Class - Case 1

$J = -J^\top$, $R = R^\top \succeq 0$ constant; quadratic $H(x) \Rightarrow \nabla H(x) = Qx, \quad g(x) \in \mathbb{R}^{n\times 1}$

Consider the following dynamics $\Rightarrow \begin{cases} \dot{x}(t) = (J-R)Qx + g(x)u \\ \dot{\hat{x}}(t) = (J-R)Q\hat{x} + g(\hat{x})u + L(y-\hat{y}) \end{cases}$

Auxiliary mappings: $\gamma(x,u) = g(x)u - L\,g(x)^\top Qx, \quad \gamma(\hat{x},u) = g(\hat{x})u - L\,g(\hat{x})^\top Q\hat{x}.$

The error dynamics $\tilde{x} = x - \hat{x}$ becomes $\Rightarrow \dot{\tilde{x}} = (J-R)Q\,\tilde{x} + \left[\gamma(x(t),u(t)) - \gamma(\hat{x}(t),u(t))\right]$

Integral Representation (FTC + MVT)

$$\gamma(x(t),u(t)) - \gamma(\hat{x}(t),u(t)) = \left(\int_0^1 \frac{\partial \gamma}{\partial x}(\bar{x}(t,s),u(t))ds\right)\tilde{x}(t)$$

where

$$\bar{x}(t,s) = \hat{x}(t) + s\tilde{x}(t), \quad s \in [0,1]$$

LPV Formulation - Case 1: Polytopic Embedding

Averaged Jacobian Matrix - Mean Value Theorem

$$A_\gamma(\bar{x}(s^*,t),L) := \int_0^1 \frac{\partial \gamma}{\partial x}(\hat{x}(t)+s\tilde{x}(t),u(t))\,ds \quad \text{with} \quad s^* \in [0,1]$$

The error dynamics becomes: $\dot{\tilde{x}}(t) = (A_0 + A_\gamma(\bar{x}(s^*,t),L))\,\tilde{x}(t), \quad A_0 := (J-R)Q$

Operating domain: Let $\mathcal{D} \subseteq \mathcal{R}^n$ be a compact set in which γ is locally Lipschitz such that $x(t), \hat{x}(t) \in \mathcal{D}$, and $u(t) \in \mathcal{U}$.

Polytopic Embedding

Since γ is Lipschitz with respect to x, its derivative can be bounded locally

$$\underline{a}_{ij} \leq \left[\frac{\partial \gamma}{\partial x}(x,u)\right]_{ij} \leq \overline{a}_{ij} \quad \forall (x,u) \in \mathcal{D} \times \mathcal{U}$$

LPV Formulation - Case 1: From Bounds to Polytope

Polytopic Embedding (continue)

There exists a finite set of **constant vertex matrices** $\{A_{\gamma,1}(L), \ldots, A_{\gamma,N_v}(L)\}$ with

$$N_v = 2^m \quad (m = \text{number of nonlinearly varying entries in Jacobian})$$

such that

$$A_\gamma((\hat{x}(t), u(t)), L) \in \mathrm{Co}\{A_{\gamma,1}(L), \ldots, A_{\gamma,N_v}(L)\}, \quad \forall (x, u) \in \mathcal{D} \times \mathcal{U}$$

Convex Combination Representation

There exist nonnegative weighting functions $h_i(\hat{x}(t), u(t)) \geq 0$ satisfying $\sum_{i=1}^{N_v} h_i = 1$ such that:

$$\boxed{A_\gamma((\hat{x}(t), u(t)), L) = \sum_{i=1}^{N_v} h_i(\hat{x}(t), u(t)) A_{\gamma,i}(L)}$$

where $A_{\gamma,i}(L)$ are **constant** (time-independent), $h_i(\hat{x}, u)$ are time-varying weights

LPV Formulation - Case 1: LMI Synthesis - Quadratic Stability

Polytopic LPV error dynamics: $\dot{\tilde{x}}(t) = \sum_{i=1}^{N_v} h_i(\hat{x}(t), u(t)) \underbrace{A_i(L)}_{A_0 + A_{\gamma,i}(L)} \tilde{x}(t), \quad A_i(L) = \bar{A}_i - L\bar{C}_i$

LMI Conditions for contraction base-approach

If $\exists\, P = P^\top \succ 0$, K such that:

$$\bar{A}_i^\top P + P\bar{A}_i - \bar{C}_i^\top K^\top - K\bar{C}_i \prec -2\lambda P, \quad \forall i = 1, \dots, N_v$$

then $L = P^{-1}K$ ensures exponential stability with rate λ and Lyapunov function $V(\tilde{x}) = \tilde{x}^\top P \tilde{x}$

Gain-Scheduled Observer (Reduced Conservatism)

If constant gain L fails or yields poor performance, compute **vertex-specific gains** $\{L_1, \dots, L_{N_v}\}$ and schedule:

$$L(\hat{x}(t), u(t)) = \sum_{i=1}^{N_v} h_i(\hat{x}(t), u(t)) L_i \quad \Rightarrow \quad \text{LMIs satisfied for larger } \lambda$$

LPV Formulation - Case 2: Nonlinear Hamiltonian Gradient

System Class - Case 2

$J = -J^\top$, $R = R^\top \succeq 0$, g constant; **non-quadratic** Hamiltonian $H(x)$

Error dynamics:

$$\dot{\tilde{x}} = (J - R - g^\top L)[\nabla H(x) - \nabla H(\hat{x})]$$

Integral Representation via Hessian (FTC + MVT)

By Mean Value Theorem:

$$\nabla H(x) - \nabla H(\hat{x}) = \left(\int_0^1 \nabla^2 H(\hat{x} + s\tilde{x}) ds \right) \tilde{x} := A_H(\bar{x}(s^*, t))\tilde{x}$$

Key insight: The same polytopic procedure from Case 1 applies here!

Bounds on Hessian entries $\nabla^2 H \Rightarrow$ vertex matrices $\{A_{H,1}, \ldots, A_{H,N_v}\}$

Results: Case 1 - Mechanical System (position)

Parameters: $m = 1 \cdot \text{kg}, \quad K = 10^3 \cdot \text{N/m}, \quad \eta = 50 \cdot \text{N s/m}, \quad \varepsilon = 2.8 \cdot \text{F/m}, \quad q_0 = 10^{-3} \cdot \text{m}.$

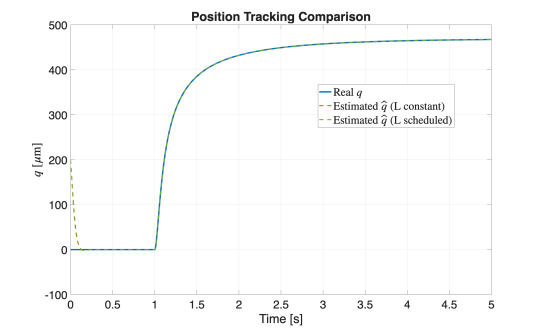


Figure 4: Position tracking

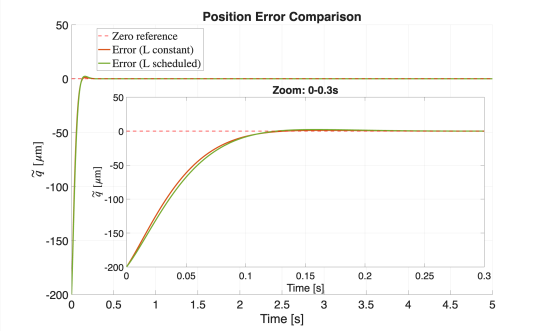


Figure 5: Position error

Results: Case 1 - Mechanical System (momenta)

Parameters: $m = 1 \cdot \text{kg}, \quad K = 10^3 \cdot \text{N/m}, \quad \eta = 50 \cdot \text{N s/m}, \quad \varepsilon = 2.8 \cdot \text{F/m}, \quad q_0 = 10^{-3} \cdot \text{m}.$

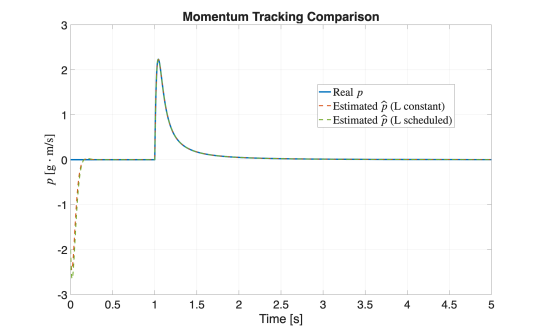


Figure 6: Momenta tracking

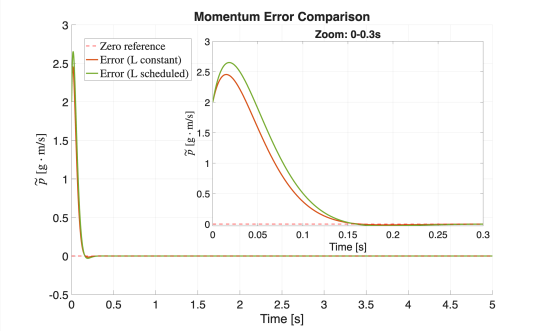


Figure 7: Momenta error

Application to HASEL Actuators

HASEL Actuator [2]

Model Complexity (1-Link):

5 states:

$x = [\theta, \ell_p, p, Q_1, Q_2]^\top$

Simultaneous nonlinearities:

- $J(\theta, \ell_p)$
- $\nabla H(\theta, \ell_p)$
- $g(\theta) = \bar{R}_0 \gamma_1 \cos(\gamma_2 \theta)$

Hamiltonian:

$$H(\theta, \ell_p, p, Q_1, Q_2) = \underbrace{H_\theta(\theta)}_{\text{Torsional spring}} + \underbrace{H_{\ell_p}(\ell_p)}_{\text{Linear spring}} + \underbrace{H_g(\theta)}_{\text{Gravity}} + \underbrace{H_p(p)}_{\text{Kinetic}} + \underbrace{H_Q(Q_1, Q_2, \theta, \ell_p)}_{\text{Electrical}}$$

⚠ COMPUTATIONAL CHALLENGE

Naive approach: $5 \times 5 = 25$ parameters

$\Rightarrow 2^{25} \approx 33$ million vertices $\Rightarrow$ **INTRACTABLE!**

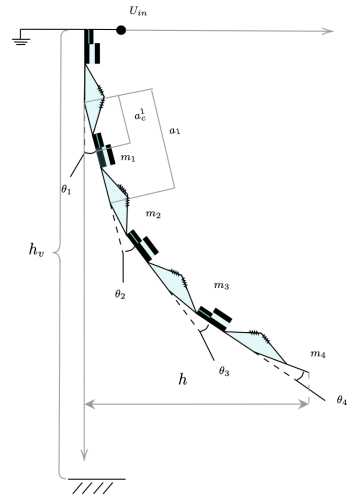


Figure 8: HASEL schematic

HASEL: Observability & Scheduling Reduction

Observability Analysis

Configuration 1: y_{pH} + θ measurement

$\Rightarrow$ Improve the conditioning, since some state are weakly observable

Configuration 2: Only y_{pH} + P scheduling

$\Rightarrow$ better validation of the observer, maintenance of the pH structure

Structural Analysis

From full Jacobian $\frac{\partial f}{\partial x}$ (25 entries):

- Identify zeros, constants
- Identify functional dependencies

Reduced Scheduling Vector

$$z = \left[\frac{1}{d(\theta, \ell_p)}, \frac{1}{C_2(\theta, \ell_p)}, \cos(\gamma_2\theta), u\right]$$

Operating domain:

- z_1, z_2: bounded by compact (θ, ℓ_p) domain
- $z_3 \in [\cos(\gamma_2\theta_{\max}), \cos(\gamma_2\theta_{\min})]$
- $z_4 \in [-8, 8]$ kV

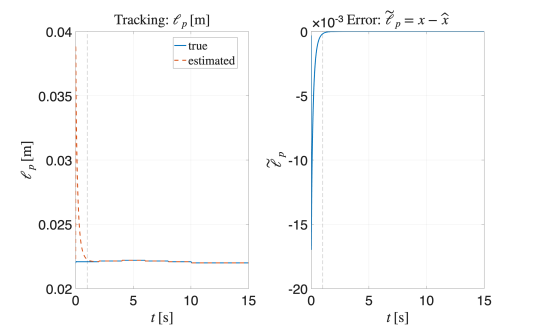
$N_v = 2^4 = 16$ vertices

Impact: From 33 million (impossible) to 16 vertices (tractable) via structural analysis

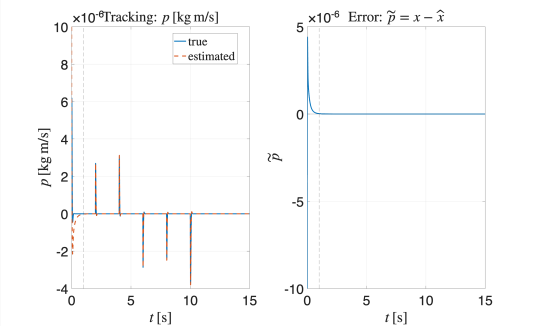
HASEL Results: Config 1 - State Tracking (1/3)

Setup: y_{pH} (charge) + θ (angle) measured, Gain-scheduled $L(\hat{x}, u)$.

Input: Staircase voltage $u \in [-8, 8]$ kV.



(a) film length ℓ_p tracking and error

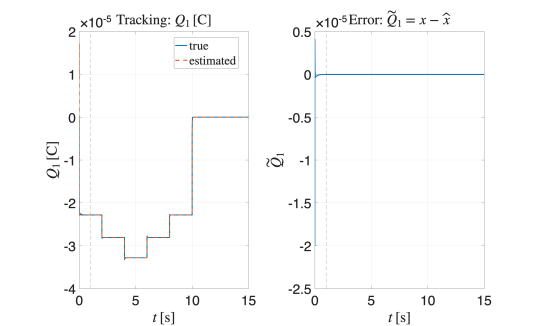


(b) Momentum p tracking and error

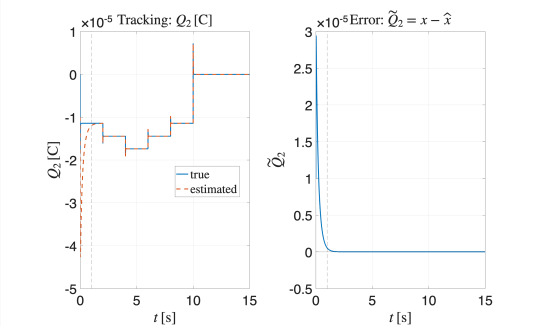
HASEL Results: Config 1 - State Tracking (2/3)

Setup: y_{pH} (charge) + θ (angle) measured, Gain-scheduled $L(\hat{x}, u)$.

Input: Staircase voltage $u \in [-8, 8]$ kV.



(a) Charge Q_1 tracking and error

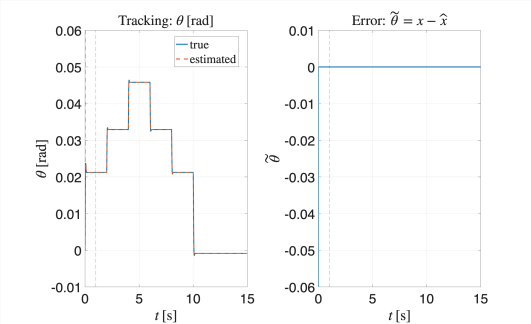


(b) Charge Q_2 tracking and error

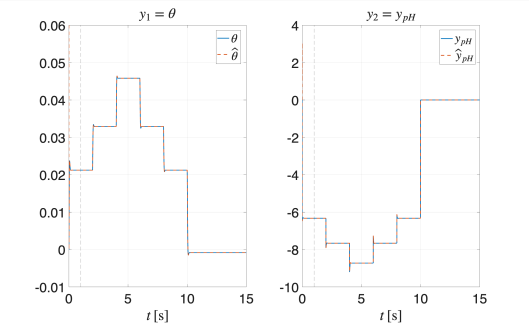
HASEL Results: Config 1 - Errors & Performance (3/3)

Setup: y_{pH} (charge) + θ (angle) measured, Gain-scheduled $L(\hat{x}, u)$.

Input: Staircase voltage $u \in [-8, 8]$ kV.



(a) Angle θ tracking and error

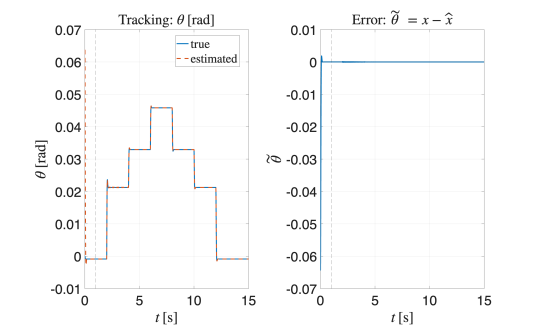


(b) Measurement tracking

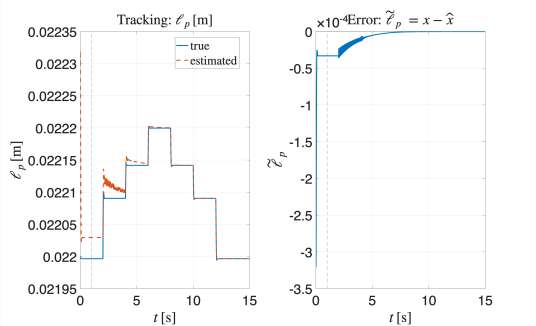
HASEL Results: Config 2 - State Tracking (1/3)

Setup: Only y_{pH} measured, Gain-scheduling also on P (reduced conservatism).

Input: Staircase voltage $u \in [-8, 8]$ kV



(a) Angle θ tracking and error

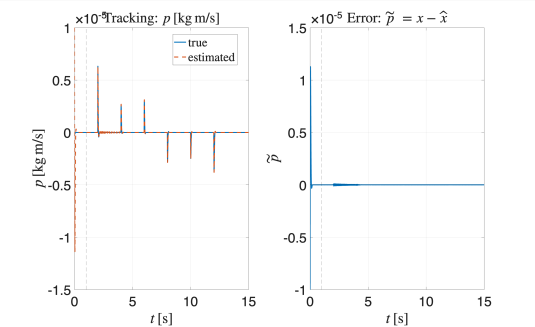


(b) film length ℓ_p tracking and error

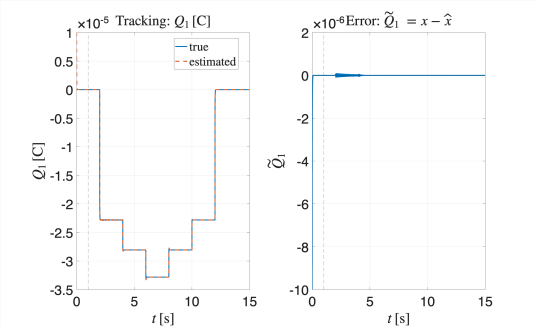
HASEL Results: Config 2 - State Tracking (2/3)

Setup: Only y_{pH} measured, Gain-scheduling also on P (reduced conservatism).

Input: Staircase voltage $u \in [-8, 8]$ kV.



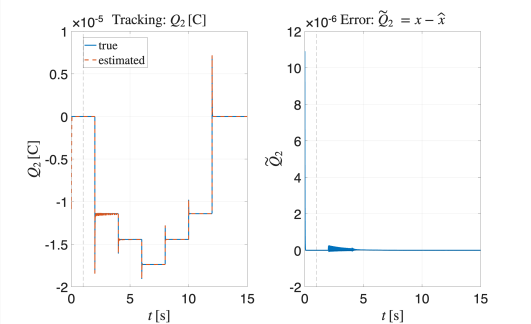
(a) Momentum p tracking and error



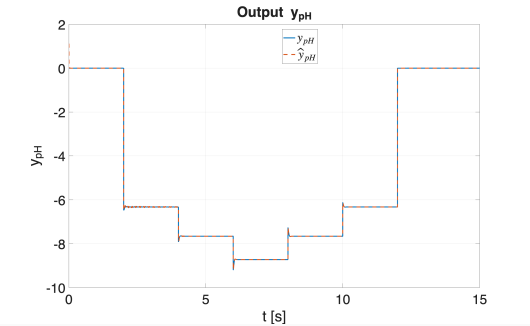
(b) Charge Q_1 tracking and error

HASEL Results: Config 2 - Errors & Observations (3/3)

Setup: Only y_{pH} measured, Gain-scheduling also on P (reduced conservatism).



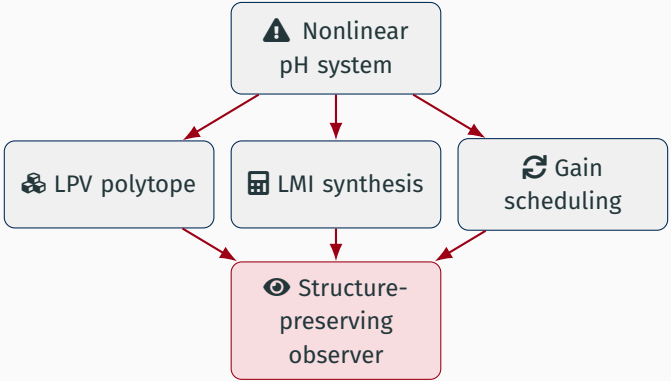
(a) Charge Q_2 tracking and error



(b) Measurement tracking

Summary and Outlook

Systematic framework: Contraction theory + LPV polytopic embedding



Validation

Mechanical (Case 1),
Electromech (Case 2),
HASEL (Complex case).

Next

MEMS, DEA, Levitation ball
Experimental validation

Soft robotics • Medical • Energy

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Thank you for your attention!

Questions?